

RAPID BEAMLINE DIAGNOSTICS FOR UPCOMING BESSY II+ SoTeXs BEAMLINE

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Abstract

As part of the BESSY II+ upgrade, the new SoTeXs (Soft-to-Tender X-ray Spectroscopy) beamline will enable high-precision, high-throughput studies of battery materials in the 0.5 keV to 5 keV energy range. At the endstation, battery cells with varying material combinations will undergo charging and discharging phases while being exposed to the beam. To ensure that, variations in the measurements are attributable to changes within the cells rather than fluctuations in beam properties, a rapid diagnostics procedure will be implemented. This procedure will monitor beam performance in between the battery measurements. This includes measurement of key parameters such as photon flux, energy resolution, and beam focus. The system combines a retractable ionization chamber for energy resolution measurements and a camera-based setup using OpenCV and ChArUco markers for determining beam spot size and position. These tools allow beam performance monitoring between sample loading cycles and represent an advance over commissioning-only diagnostics on current BESSY II beamlines. This paper presents the technical requirements of the SoTeXs beamline and a selection of potential diagnostic tools.

INTRODUCTION

BESSY II+ is the latest major upgrade to the BESSY II synchrotron light source at Helmholtz-Zentrum Berlin (HZB), developed to meet the growing demands of next-generation research in energy materials [1].

At the core of this upgrade is SoTeXs (Soft to Tender X-ray Spectroscopy), a flagship beamline tailored for operando and in situ battery research. Investigating electrochemical processes in real time requires a stable, high-flux photon source with excellent energy resolution and fast scanning capabilities.

SoTeXs is designed to deliver rapid energy scans in the 0.5 keV to 5 keV range, enabling high-throughput measurements across relevant absorption edges in modern battery chemistries. In collaboration with the National Metrology Institut (Physikalisch-Technische Bundesanstalt - PTB), the beamline also serves as a metrology platform, incorporating rapid beam diagnostics to minimize measurement uncertainty and improve reproducibility.

This combination of advanced spectroscopy and precision metrology makes SoTeXs a central component of the BESSY II+ upgrade. SoTeXs is still in an early stage of the

development. The first light is planned for 2029. The following paper will present the current state of the design with the focus on methods to perform rapid beamline diagnostics.

GENERAL DESCRIPTION OF THE BEAMLINE

The optical layout of the SoTeXs beamline, as shown in Fig. 1, is specifically designed to deliver high photon flux and spectral resolution in the tender X-ray regime from 0.5 keV to 5 keV. At the SoTeXs endstation, a dedicated battery-cell storage unit will provide a cyclical flow of samples.

The source of the beamline is a newly developed planar cryogenic permanent magnet undulator [2], optimized for the soft X-ray range, with higher harmonics extending into the tender X-ray range. The emitted radiation is directed onto a toroidal mirror (M1) at an incidence angle of 0.7° . This first mirror provides both horizontal focusing and vertical collimation of the undulator radiation and reflects the light onto the pre-mirror (M2) of the monochromator. The monochromator is equipped with two gratings with line densities of $1200\text{ l}\cdot\text{mm}^{-1}$ and $2400\text{ l}\cdot\text{mm}^{-1}$. Depending on the experiment, it is possible to choose between high photon flux and high resolution [3].

To improve reflectivity—and therefore photon flux—in the tender X-ray range, all optics, including mirrors and gratings, are coated with multilayer coatings. Each mirror will carry different coated areas on the same substrate. By moving the mirror normal to the beam, the illuminated region (and therefore the coating) is switched. This approach benefits from experience at ELISA [4].

After the grating, the beam is focused by a cylindrical mirror (M3)—again at an incidence angle of 0.7° —on the exit slit. The slit defines the energy resolution and spatial coherence of the beam delivered to the experiment.

Beyond the slit, the beam is focused onto the sample by a Kirkpatrick–Baez (KB) mirror system, enabling small spot sizes suitable for micro-spectroscopy.

This layout provides the flexibility to perform both fixed-energy and energy-scanning experiments with high spatial and spectral resolution (with emphasis on energy scanning). The beamline is fully compatible with in situ and operando battery setups and incorporates integrated beam-diagnostics infrastructure for rapid monitoring and calibration.

ENABLING TECHNOLOGIES FOR SoTeXs

To meet the scientific needs of the SoTeXs beamline—particularly the operando analysis of modern battery materials—several key technological innovations are being implemented. These are tailored to provide high photon flux, fast

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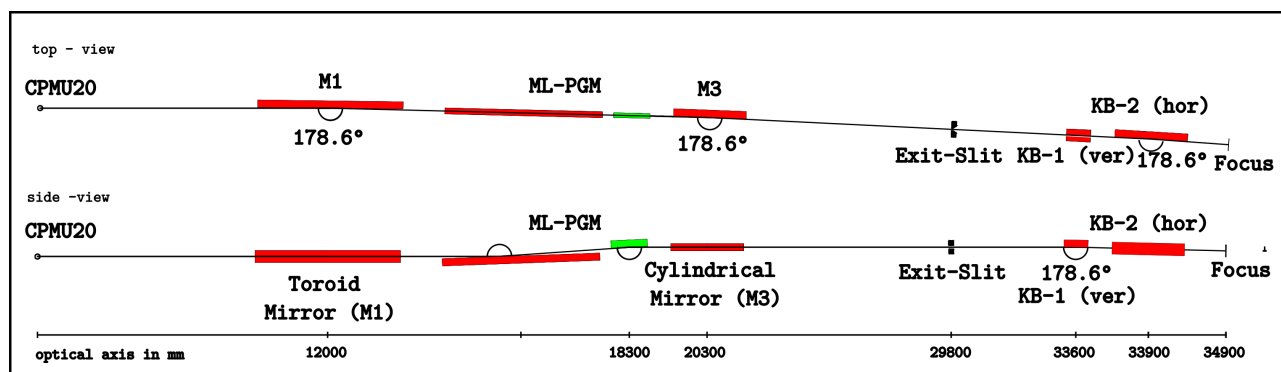


Figure 1: The optical path of SoTeXS (export from ray-tracing software RAY-UI).

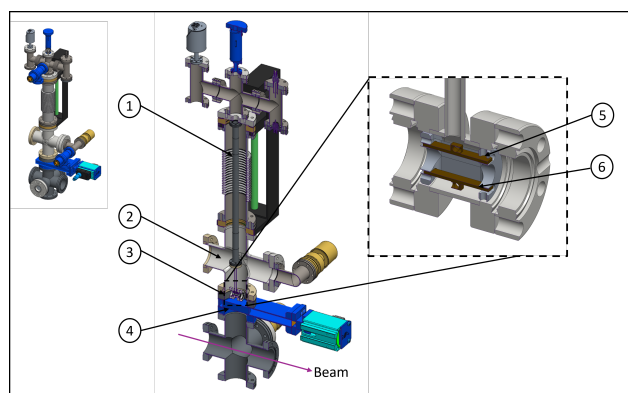


Figure 2: Schematic of the ionization chamber; 1. Linear drive system; 2. T-section; 3. Ionization chamber; 4. Disc valve; 5. Electrode A; 6. Electrode B.

energy scanning, and precise beam diagnostics within the tender X-ray regime 0.5 keV to 5 keV, a spectral region of increasing importance for light element chemistry.

Ionization Chamber for Fast Beam Energy Resolution Measurement

Beamlines are typically equipped with a fixed segment containing an ionization chamber. These chambers operate similarly to capacitors: two electrodes are placed inside a gas volume, and when the X-ray beam passes through, the gas becomes ionized. This ionization generates a measurable current at the electrodes, which is proportional to the incident photon flux and also dependent on the electron configuration of the gas and the energy of the radiation. Therefore, ionization chambers allow us not only to measure the photon flux, but also to assess the energy resolution of the radiation coming from the monochromator.

The aim for SoTeXS is to perform quick energy-resolution scans between sample exchange phases. To support this, a compact, movable ionization chamber has been developed. The system consists of a small gas-filled cavity equipped with two electrodes. A linear drive system enables movement in and out of the beam.

Figure 2 shows the full system, with a detailed view of the chamber.

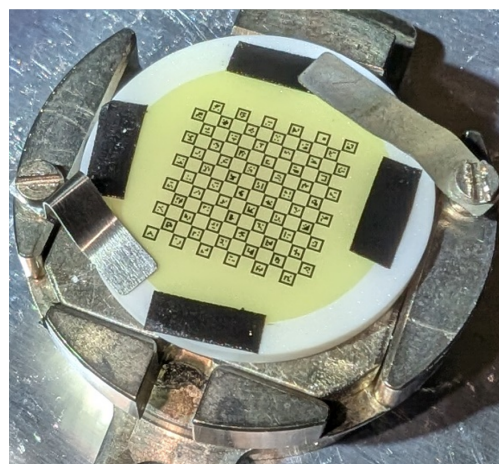


Figure 3: ChArCO-Code on fluorescence screen

This small chamber can be inserted into the beam path when needed and retracted afterward. The first T-bar section can be used for maintenance of the gas cell. The key advantage of this design is that it eliminates the need to re-pump the chamber to ultra-high vacuum (UHV) each time, significantly reducing downtime and increasing operational efficiency.

ChArUco Codes for Fast Beam Position and Focus Size Measurement

At previous BESSY II beamlines, the beam focus position was typically measured and aligned during initial commissioning. However, for reproducible datasets, especially in long duration battery measurements, it is important to maintain the same beam profile and focus position. In addition to increasing the quality of the measurements with the beamline, regular beam profile and focus position measurements enable the tracking of slow changes in the beamline, which can be used for preventative maintenance. To address this, a system based on a fluorescence screen with a printed ChArUco code, which was initially tested and further developed at the EMIL beamline [5], will be implemented. A ChArUco pattern is a combination of ArUco codes arranged in a chessboard formation Fig. 3. This provides a robust set of reference points that enables a script running

with OpenCV to perform homography on a tilted image. The camera is controlled via EPICS-AreaDetector, so this system can capture the beam, calculate distortions, and measure the beam size and position. The camera is mounted and focused on the pattern. A calibration image of the pattern under visible light is then taken. Using the detected distortion, the software computes a rectified view normal to the screen. The system is now calibrated and ready to measure beam size and position. A more detailed description of the process can be found in Smith et al. [5].

Placing multiple of these systems at points of interest along the beamline (e.g. the intermediate focus of M3) allows not only measurement but also traceability of potential errors. Deviations of focus size and beam position between the real-world beamline and the simulated version in the ray-tracing program RAY-UI enables the diagnosis of potentially misaligned optics and allows to optimize the performance of the SoTeXS beamline.

Next-Generation Monochromator and Multilayer Optics

SoTeXS is designed to extend deeper into the tender X-ray range than previous beamlines at BESSY II. In this energy range, single-layer optics suffer strong absorption and low grazing-angle reflectivity, leading to reduced efficiency in classical plane mirror monochromators [3, 6].

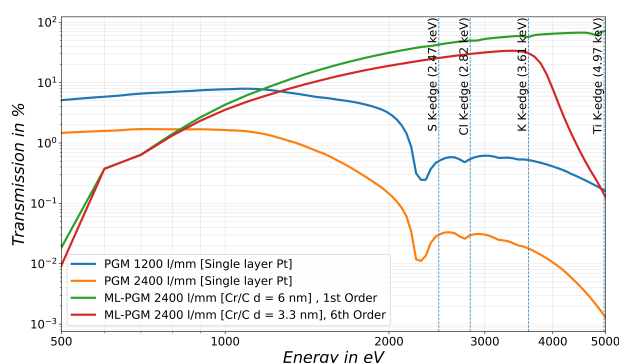


Figure 4: A comparison of the theoretical transmission between multilayered and normal PGMs.

Figure 4 shows theoretical possible transmission of different monochromator systems, with an selection of the absorption edges of important battery materials. Using the classical PGM-Design leads in combination with a single layer coating (here Pt) to a substantial decline of transmission from approx. 2.3 keV.

By employing custom-designed multilayer-coated mirrors and gratings, SoTeXS will significantly enhance photon throughput and extend usable bandwidth, while maintaining beam quality and stability across the entire energy range [7]. Initial tests at U41-TXM beamline at BESSY II have demonstrated promising performance, validating the feasibility of this approach for SoTeXS [8, 9].

A critical component of the beamline is the monochromator, which must deliver both high energy resolution and fast scanning capabilities. The selection of a suitable design is currently under investigation, as the requirements for mechanical stability, scanning speed, and repeatability exceed those of existing monochromator systems in this energy range.

The combination of custom-engineered multilayer optics and a next-generation monochromator will be key to fulfilling the beamline's mission: delivering high photon flux with fast, energy-resolved access across a wide spectral range, optimized for battery science.

FUTHER CHALLENGES ON THE ROAD TO SoTeXS

SoTeXS is being developed as a dedicated beamline for high-throughput, high-precision X-ray studies of battery materials. Its optical design, fast energy scanning, and integrated diagnostics aim to meet the demands of modern operando experiments in the tender X-ray range.

However, several technical challenges remain. The final design of the monochromator must balance scanning speed, beam stability, and energy resolution—requirements that exceed those of conventional systems. Although the implementation of multilayer coatings across all optics appears promising, experience from other beamlines (e.g. ELISA) will reveal its true potential. Reliable in-situ diagnostics, including the CharuCo-based beam monitoring and movable ionization chambers, must be seamlessly integrated into the experimental workflow without compromising vacuum integrity or alignment stability.

Addressing these challenges will be essential to realize the full potential of SoTeXS as a flagship beamline for BESSY II+, capable of delivering reproducible, metrology-grade data for battery materials.

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REFERENCES

- [1] BESSY II+ Upgrade, https://www.helmholtz-berlin.de/forschung/quellen/bessy-2-plus-upgrade_en.html
- [2] S. Schäfer *et al.*, "A new cryogenic permanent magnet undulator at BESSY-II: the CPMU-20", in *Proc. IPAC'24*, Nashville, TN, May 2024, pp. 3773-3776.
doi:10.18429/JACoW-IPAC2024-THPS20
- [3] R. Follath, "The versatility of collimated plane grating monochromators", *Nucl. Instrum. Methods Phys. Res. A*, vol.

467–468, pp. 418–425, Jul. 2001.

[doi:10.1016/S0168-9002\(01\)00338-2](https://doi.org/10.1016/S0168-9002(01)00338-2)

- [4] S. Vadilonga *et al.*, “Optical layout and endstation concept for the Enhanced Liquid Interface Spectroscopy and Analysis (ELISA) beamline at BESSY-II”, *Synchrotron Radiat. News*, vol. 35, no. 3, pp. 67–72, Jun. 2022.

[doi:10.1080/08940886.2022.2082213](https://doi.org/10.1080/08940886.2022.2082213)

- [5] W. Smith *et al.*, “Using ArUco codes for beam spot analysis with a camera at an unknown position”, in *Proc. ICALEPCS'23*, Cape Town, South Africa, Oct. 2023, pp. 1264–1268.

[doi:10.18429/JACoW-ICAEPSCS2023-THMBCM030](https://doi.org/10.18429/JACoW-ICAEPSCS2023-THMBCM030)

- [6] X. Yang *et al.*, “Design of a multilayer-based collimated plane-grating monochromator for tender X-ray range”, *J. Synchrotron Rad.*, vol. 24, pp. 168–174, Jan. 2017.

[doi:10.1107/S1600577516017884](https://doi.org/10.1107/S1600577516017884)

- [7] A. Sokolov *et al.*, “Optimized highly efficient multilayer-coated blazed gratings for the tender X-ray region”, *Opt. Express*, vol. 27, no. 12, pp. 16833–16846, 2019.

[doi:10.1364/OE.27.016833](https://doi.org/10.1364/OE.27.016833)

- [8] S. Werner *et al.*, “Spectromicroscopy of nanoscale materials in the tender X-Ray regime enabled by a high efficient multilayer-based grating monochromator”, *Small Methods*, vol. 7, no. 1, p. 2201382, Jan. 2023.

[doi:10.1002/smt.202201382](https://doi.org/10.1002/smt.202201382)

- [9] Multilayer-coated diffraction gratings for high transmission tender and soft X-ray energy range monochromators, https://www.helmholtz-berlin.de/forschung/oe/wi/optik-strahlrohre/groups/metrology1/multilayer-coated-gratings_en.html